

LIRUI WANG

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EDUCATION

University of Washington, Seattle, WA

Sep 2016 - Mar 2021

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

Master of Science, Computer Science, College of Arts and Sciences

GPA: 3.96/4.00, *summa cum laude*

Advisor: Professor Dieter Fox

RESEARCH PROJECTS

6D Instance Grasping with Point Cloud

Jan 2020 - Sep 2020

- Augmented reinforcement learning with planner demonstrations and goal auxiliary tasks, for learning closed-loop control policies in 6D robotic grasping using point clouds.
- Achieved over 90% success rates on grasping various ShapeNet objects and YCB objects in simulation.

Joint Motion and Grasp Planning

Mar 2019 - Dec 2019

- Proposed an optimization-based integrated planner that optimizes manipulation trajectory with on-line grasp synthesis and selection.
- Demonstrated robust and efficient motion plans for grasping in cluttered scenes in the real world and outperformed several baselines.

Pose Tracking with the Deep Iterative Matching Network

Jun 2018 - Mar 2019

- Evaluated DeepIM on YCB dataset in depth and improve training on synthetic data by generating diverse pose distribution and realistic occlusions.
- Extended DeepIM to achieve state-of-the-art pose tracking accuracy and AR applications.

Unsupervised Flow and Depth Methods Study

Mar 2018 - Jun 2018

- Investigated learning methods for 3D geometry tasks such as depth, flow, and structure from motion.
- Explored self-supervised structures works such as SfMlearner and MonoDepth and analyzed them extensively on a static scene manipulation setting.

Baxter Hand-Eye Coordination

Sep 2017 - Mar 2018

- Developed software for the Baxter robot to grasp and manipulate daily objects detected by PoseCNN.
- Studied and implemented camera models, hand-eye calibrations, robot kinematics, computer graphics, and ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA

Jun 2020 - Oct 2020

Applied Research Intern, NVIDIA

- Investigated model-based RL for autonomous driving with a learned sequential traffic predictive model with teacher-aided explorations.

- Used the predictive model latent as the policy input for behaviors such as cut-in avoidance and adopt the value function for long-term planning.

Seattle, WA

Sep 2019 - Present

Teaching Assistant

- Assisted teaching for computer vision, deep learning, and artificial intelligence courses.
- Held office hours, helped homework grading, and answered questions on the discussion board.

Shenzhen, China

Jun 2017 - Aug 2017

Robotics Engineer Intern, DJI Technology

- Implemented drone functionality such as take-off, tag detection, and cooperation with vehicles.
- Applied SLAM to mapping and localization, and built a state machine to generate high-level plans.

LEADERSHIP AND INVOLVEMENT

Seattle, WA

Sep 2016 - Feb 2018

ARUW Computer Vision Lead

- Managed the a team of 10 students at UW to design computer vision algorithms for an autonomous robot battle, including vehicles with onboard NVIDIA TX1 and drones with DJI Manifold with Guidance, in the RoboMasters competition hosted by DJI.
- Performed system integration of feedback controls, navigation systems and serial communication.

MISCELLANEOUS PROJECTS

- **Spatial Audio Processing in the SoundScape System:** Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle.
- **Sheet Music Editor:** Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded 1st place out of 35 teams in a Seattle Hackathon.
- **Embedded Drowsiness Detection System:** Built an embedded system on Raspberry Pi that that simulates a real-time drowsiness detection system based on human face detection.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification:** Investigated example reweighting and adversarial training techniques to robustify a fossil classification task and surveyed on attacks and defense of adversarial examples for neural networks.
- **Continuous Trajectory Optimization with Machine Learning Techniques:** Summarized the formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference on a factor graph using Gaussian Process.
- **Microsoft Server Mover Project:** Built an autonomous mobile manipulator, a UR5 arm mounted on a server forklift, to pull out servers from server rack and transport it to repair.

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

PUBLICATIONS

- **Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds**
Lirui Wang, Yu Xiang, Dieter Fox
Under Review

- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**

Lirui Wang, Yu Xiang, Dieter Fox

In Proceedings of Robotics: Science and Systems (RSS 2020)

SELECTED COURSES

- **Computer Science:** Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Mobile Robots, Robustness in Machine Learning, Interactive Learning, Probabilistic Robotics, Reinforcement Learning
- **Electrical Engineering:** Digital Signal Processing and Filtering, Control Theory, Embedded System, Discrete Time Linear System, Systems, Controls, and Robotics Capstone, Optimization and Learning for Control
- **Math:** Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis, High Dimensional Statistics and Statistical Learning, Statistical Learning: Modeling, Prediction, And Computing
- **English:** Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Systems:** Arduino, NVIDIA TX Series, STM32, Franka Panda, Baxter, UR5e, DJI Manifold
- **Languages:** Python, C++/C, Matlab, Java
- **Tools:** Git, CMake, OpenCV, Tensorflow, Pytorch, Latex, OpenGL, ROS, Ubuntu
- **Hardware:** 3D Printing, Laser Cutting, Machining, Soldering